

Task 5:

1) Declare an auto Variable inside a block and prints its Value

```
#include <stdio.h>
```

```
int main() {
```

```
    printf("Inside main block: \n"),
```

```
    {
```

```
        auto int x = 10;
```

```
        printf("auto variable x = %d \n", x);
```

```
    }
```

```
    return 0;
```

```
}
```

output:

Inside main block:

auto variable x = 10

2) Use a register Variable in a for loop to count from 0 to 4 and print the result

```
#include <stdio.h>
```

```
int main () {
```

```
    register int i;
```

```
    for (i = 0; i < 5; i++) {
```

```
        printf("Register variable i = %d \n", i);
```

```
    }
```

```
    return 0;
```

```
}
```

output:

Register variable i = 0

Register variable i = 1

Register variable i = 2

Register variable i = 3

Register variable i = 4

3) Static Variable in function

```
#include <stdio.h>
```

```
void counter() {
```

```
    static int count = 0;
```

```
    count++;
```

```
    printf("Function called %d times\n", count);
```

```
}
```

```
int main() {
```

```
    counter();
```

```
    counter();
```

```
    counter();
```

```
    return 0;
```

```
}
```

output:

Function called 1 times

Function called 2 times

Function called 3 times

4) Extern Variable across files

file1.c

```
#include <stdio.h>
```

```
int count = 5;
```

```
void display() {
```

```
    printf("Extern Variable count = %d\n", count);
```

```
}
```

file2.c

```
#include <stdio.h>
```

```
extern int count;
```

```
void display();
```

```

int main() {
    display();
    return 0;
}

```

compile:

```

gcc file1.c file2.c -o extern-demo
./extern-demo

```

output:

Extern Variable Count = 5

5) Mixing auto and static Variables

```

#include <stdio.h>

```

```

void demo() {

```

```

    auto int a = 0;

```

```

    static int s = 0;

```

```

    a++;

```

```

    s++;

```

```

    printf("Auto = %d, static = %d\n", a, s);
}

```

```

}

```

```

int main() {

```

```

    demo();

```

```

    demo();

```

```

    demo();

```

```

    return 0;
}

```

```

}

```

output:

Auto = 1, static = 1

Auto = 1, static = 2

Auto = 1, static = 3